Paper 5 Analysis Table

Category	Details
Research Paper Title	An Explainable AI-driven Deep Neural Network for Accurate Breast Cancer Detection from Histopathological and Ultrasound Images
Objective	To develop an accurate and explainable deep learning model (DNBCD) for detecting breast cancer using both ultrasound and histopathological images.
Problem Statement	Traditional diagnostic methods are complex, time-consuming, and dependent on expert interpretation, while many AI models lack interpretability (black-box issue). (Nature)
Model Used	- Deep Learning: DenseNet121 (Transfer Learning) - Custom CNN layers (Dense, Dropout, GAP) - DNBCD (proposed model)
Programming / Tools	➢ Python 3.7 ➢ TensorFlow / Keras ➢ Kaggle GPU (P100) ➢ Image preprocessing & augmentation
Dataset	- BreakHis-400x (Histopathology): 1820 images - BUSI (Ultrasound): 1578 images (Nature)
Explainability	✔ Grad-CAM used to highlight important tumor regions for decision transparency (Nature)
Pros	✔ Multi-modal data (ultrasound + histopathology) ✔ High accuracy (93.97% & 89.87%) ✔ Explainable AI (Grad-CAM) ✔ Uses transfer learning (efficient training)
Cons	✘ High computational cost ✘ Requires GPU resources ✘ Still limited real-world validation ✘ Complex architecture
Challenges	⚠ Data imbalance (handled using class weights) ⚠ Need for large labeled datasets ⚠ Model generalization ⚠ Clinical adoption barriers

Conclusion	The DNBCD model achieves high accuracy and provides interpretable results, outperforming existing methods and showing strong potential for clinical use. (Nature)
Critical Comment	This paper is stronger than previous ones because it uses **multi-modal data + explainability**, making it closer to real clinical applications. However, it still lacks segmentation and advanced XAI techniques. Your FYP (UltraXAI) can improve this by adding **Swin-UNet segmentation + Grad-CAM++ + SHAP**, making your system more advanced and explainable.